SIMATS ENGINEERING

ASSESSMENT TOOL 5: Technical Presentation

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1519

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 10%

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 25**

Question Count: 5

Code of Conduct

I THIRU KUMARAN .S / 192521149 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Thiru Kumaran S

Assessment Tasks

Technical Presentation Topics

Prepare a 6-8 minute presentation on any one of the following topics. Your slides must include definitions, architecture, industry relevance, and one practical example.

Here are 10 additional technical presentation topics in a similar style, suitable for a 6–8 minute presentation with definitions, architecture, industry relevance, and a practical example:

1. **Containerization and Cloud-Native Applications: Transforming Modern Software Deployment**
   - Explore containers, container architecture, benefits, and a real-world deployment example.
2. **Serverless Computing: Reducing Infrastructure Management in the Cloud**
   - Discuss Function-as-a-Service (FaaS), architecture, business advantages, and practical use cases.
3. **Edge Computing vs Cloud Computing: Choosing the Right Computing Paradigm**
   - Compare architectures, applications, and industry adoption scenarios.
4. **Microservices Architecture in Cloud Environments**
   - Explain microservices design, benefits over monolithic systems, and industry implementations.
5. **Cloud Security Fundamentals: Protecting Data and Applications**
   - Cover security models, shared responsibility architecture, and real-world security practices.
6. **Multi-Cloud and Hybrid Cloud Strategies for Enterprise IT**
   - Analyze deployment models, business benefits, challenges, and practical examples.

7. **The Role of Artificial Intelligence and Machine Learning in Cloud Platforms**
   - Discuss AI/ML cloud services, architecture, and industry applications.
8. **DevOps and Continuous Integration/Continuous Deployment (CI/CD) in Cloud Computing**
   - Explain DevOps principles, CI/CD architecture, and practical implementation examples.
9. **Cloud Storage Technologies: Object, Block, and File Storage Compared**
   - Compare storage architectures, use cases, and industry adoption patterns.
10. **Internet of Things (IoT) and Cloud Computing: Building Connected Ecosystems**
    - Examine IoT-cloud architecture, benefits, challenges, and real-world applications.
11. ***Cloud Migration Strategies: Moving Enterprise Applications to the Cloud***
    - Examine migration approaches, cloud architecture considerations, benefits, challenges, and a successful migration case study.
12. **Big Data Analytics in Cloud Computing**
    - Explore cloud-based data processing architectures, analytics platforms, business value, and real-world applications.
13. **Identity and Access Management (IAM) in Cloud Environments**
    - Discuss authentication, authorization architectures, security benefits, and practical enterprise implementations.
14. **Content Delivery Networks (CDNs) and Their Role in Cloud Services**
    - Explain CDN architecture, performance optimization, industry relevance, and examples from popular web platforms.
15. **Server Virtualization and Its Role in Cloud Computing**

    Explain virtualization architecture, hypervisors, virtual machines, resource allocation, and the benefits of virtualization in cloud data centers.

16. **Cloud Storage Technologies and Architecture**

    Discuss cloud storage concepts, storage architectures, scalability, availability, data replication, and practical applications such as Google Drive and Amazon S3.

17. ***Cloud Security: Threats and Protection Mechanisms***

    Explain common cloud security threats, data protection, encryption, network security, security controls, and practical approaches used by cloud service providers.

18. **Infrastructure as a Service (IaaS): Architecture and Applications**

    Explain the IaaS model, virtual computing resources, storage and networking services, advantages, limitations, and practical enterprise applications.

19. **Platform as a Service (PaaS) for Cloud Application Development**

    Discuss PaaS architecture, development environments, application deployment, scalability, advantages, limitations, and examples of PaaS platforms.

20. **Software as a Service (SaaS): Architecture and Business Applications**

    Explain SaaS architecture, features, advantages, subscription-based software delivery, security considerations, and examples such as Google Workspace and Microsoft 365.

Minimum slide flow:

- Title
- Aim and objective
- Core technical explanation
- Architecture or process diagram
- Methodology
- Real-world use case
- Conclusion and key takeaway

Technical Presentation Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Technical Content	30%	Content is highly accurate, relevant, and insightful	Content is correct with minor gaps	Basic content with limited depth	Content is inaccurate or superficial
Structure and Flow	20%	Excellent logical flow and slide organization	Good structure with minor issues	Some organization but weak flow	Disorganized presentation
Use of Visuals	15%	Visuals strongly support technical communication	Relevant visuals used well	Limited or unclear visuals	No meaningful visuals
Communication Skills	20%	Confident, clear, and engaging delivery	Clear delivery with minor hesitation	Moderate clarity but uneven delivery	Weak delivery and low clarity
Professionalism	15%	Well-prepared and audience-focused presentation	Mostly professional	Basic preparation visible	Poor preparation and professionalism

TOPIC: **INFRSTRUCTURE AS A SERVICE (IaaS)**

INTRODUCTION:

- **Meaning:** IaaS means Infrastructure as a Service.
- **Service:** It provides IT resources through the internet.
- **Rent:** Users can rent servers, storage and networks.
- **Benefit:** No need to buy physical hardware.
- **Scaling:** Resources can be increased or decreased easily.
- **Examples:** AWS, Microsoft Azure, Google Cloud.

OBJECTIVES:

- Learn the IaaS model.
- Understand virtual machines.
- Learn storage and networking.
- Know advantages and limitations.
- Study real-world applications.

CORE TECHNICAL EXPLANATION:

- **Meaning:** IaaS means Infrastructure as a Service.
- **Service:** It provides IT resources through the internet.
- **Rent:** Users can rent servers, storage and networks.
- **Benefit:** No need to buy physical hardware.
- **Scaling:** Resources can be increased or decreased easily.
- **Examples:** AWS, Microsoft Azure, Google Cloud.

ARCHITECTURE OR PROCESS DIAGRAM:

- **Step 1:** User requests resources from the cloud portal.
- **Step 2:** IaaS services give virtual machines, storage and network.
- **Step 3:** Virtualization shares the physical servers.
- **Step 4:** Applications run on these resources and can be scaled when needed.

REAL-WORLD USE CASE:

- **Sales:** More users come during sales.
- **Scale up:** Extra virtual machines are created.
- **Storage:** Storage keeps product data.
- **Traffic:** Load balancer handles traffic.
- **Scale down:** Resources are reduced after the sale.

PHOTO OF THE PRESENTATION:



CONCLUSION:

- IaaS provides IT resources through the cloud.
- *It provides compute, storage and networking.*
- It reduces hardware cost.
- It is flexible and scalable.
- It is useful for many businesses.

INFRASTRUCTURE AS A SERVICE

Infrastructure as a service (IaaS) is a cloud computing service model that provides virtualized computing resources over the Internet to allow user to access server, storage, networking and other computing maintaining Physical hardware

It IaaS the cloud service provider manages the Physical infrastructure including servers, data centre storage device and networking equipment.

The main component of IaaS include virtual machine, storage, networking, operating system, and Security service. Virtual machine provide computing Power which cloud storage store file and database.

Some Popular example of IaaS provide are Amazon web services (AWS), microsoft Azure, and google cloud platform (GCP).

In conclusion infractructure service is an important
Part of cloud computing it provide computing
infrastructure on demand reduce cost improve
scalability. and help Organization deploy applicatio
faster.